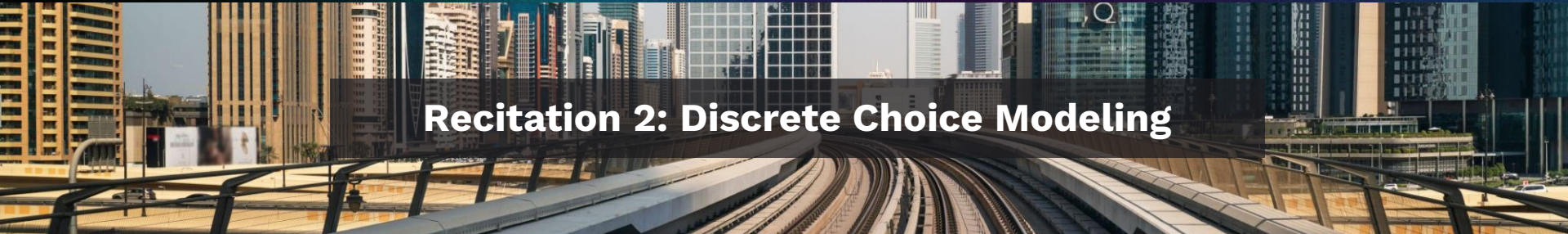


AI for Transportation: From concepts to implementation



Recitation 2: Discrete Choice Modeling

AI for Transportation

R1: Introduction

R2: Discrete Choice Modeling

R3: Real-World AI Case Studies

R4: Generative AI

AI for Transportation

R2: Discrete Choice Modeling

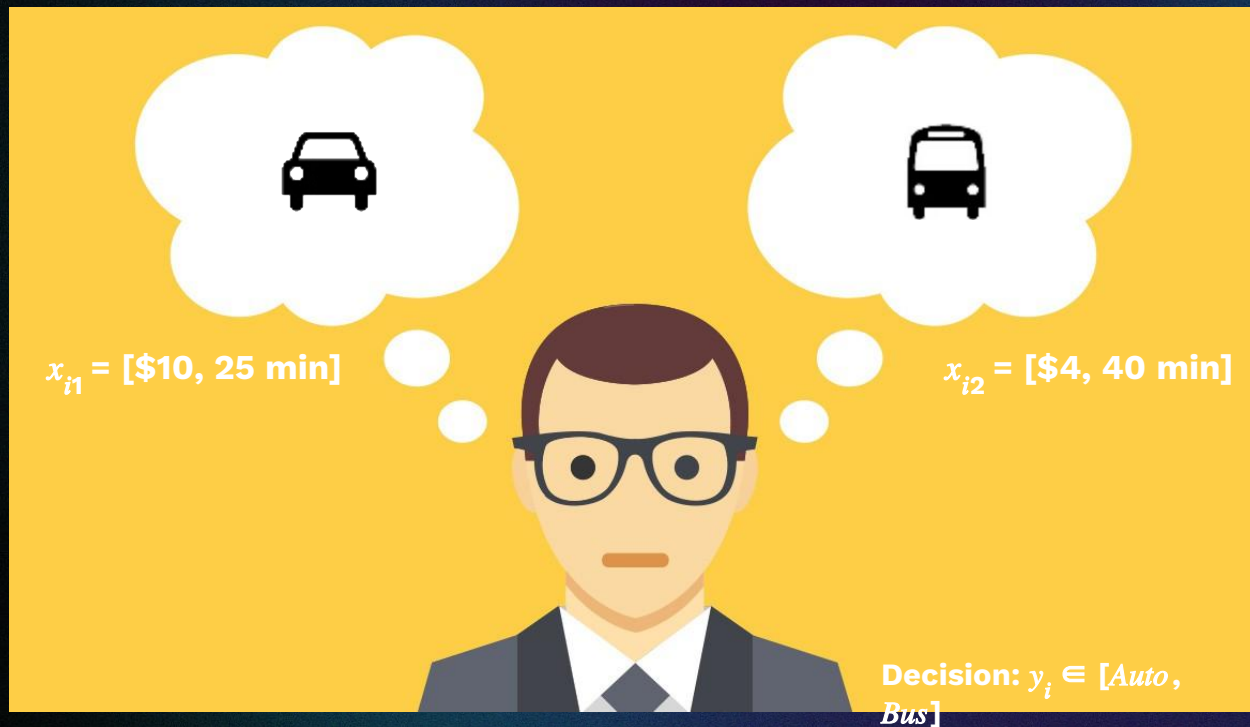
S1: Refresher on Discrete Choice Modeling

S2: Leveraging ASU-DNN

S3: Understanding the Results

S4: Future of DNN for Discrete Choice Modeling

Travel mode choice



Classical analytical framework for discrete choice

Approach for modeling individual choices from a set of feasible discrete alternatives

Decision-Maker

Individuals (person/household)

Socio-economic variables (e.g. age, gender, income)

Alternatives

Decision-maker i selects one and only one alternative from a choice set $C = \{1, 2, \dots, K\}$

Collectively exhaustive & mutually exclusive

Attributes of alternatives

e.g. price, quality, time, reliability,...

Decision rule

Utility maximization, etc.

Factors that Impact Mode Choice

- Characteristics of the trip maker (individual attributes)
- Characteristics of the trip
- Characteristics of the transportation system (attributes/performance)

Transportation System Attributes

- **Time**

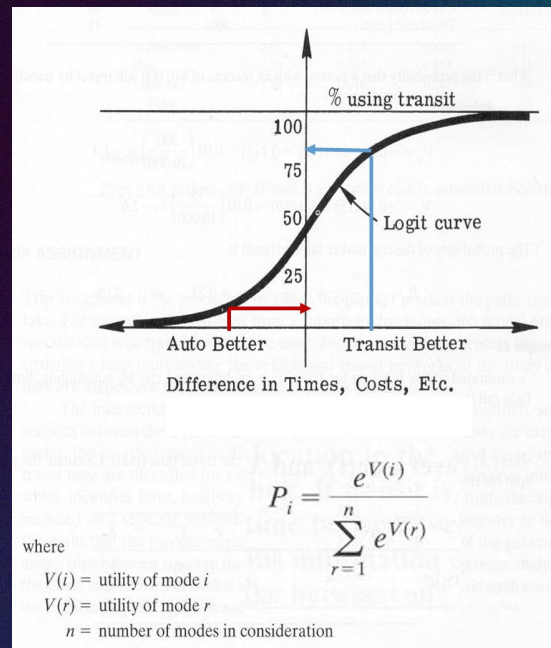
- in-vehicle (IVTT)
- out of vehicle (OVTT)
 - Automobile
 - Walk to and from car (access time), parking
 - Transit
 - Walk to and from stop (access and egress times)
 - Waiting time Frequency of service/headways
 - Connection time

- **Cost**

- Fare
- Fuel costs
- Parking fees
- Tolls

Discrete Choice Models: Logit

- Discrete choice models are based on economic theory, assuming rational consumers
- They estimate the probability of choosing a specific alternative, based on the characteristics of the individual and trade offs among the attributes of the alternatives



Random Utility Approach

- Discrete Choice Analysis

- $U_{in} = V_{in} (\text{attributes of } i, \text{ characteristics of } n, \beta) + \varepsilon_{in}$
- V_{in} = Systematic utility, a function of observable variables
- ε_{in} = error term

Random Utility Approach

- Discrete Choice Analysis

- $U_{in} = V_{in}$ (attributes of i , characteristics of n , β) + ε_{in}
- V_{in} = Systematic utility, a function of observable variables
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Assumptions about the distribution of ε lead to different “choice models” i.e., different functions for $P(i)$

Random Utility Approach

- Discrete Choice Analysis

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Assumptions about the distribution of ε lead to different “choice models” i.e., different functions for $P(i)$

- Decision Rule: maximize utility

- $P_n(i) = P(U_{in} \geq U_{jn}, \forall j \in C_n)$

C_n = set of feasible alternatives for individual n

$P_n(i)$ = probability that individual n chooses alternative i .

The Systematic Utility, V_{in}

- $V_{in} = \beta X$
 - β : vector of parameters
 - X : vector of explanatory variables
- Variables X
 - Attributes of alternatives: e.g., travel time, travel cost
 - Characteristics of decision-makers: e.g., age, gender, income, ...
 - $V(\text{bus})$ = function of (walk time, in vehicle time, fare, income..)
 - $V(\text{auto})$ = function of (travel time of auto, parking cost, ...)
- Parameters β represent tastes, which may vary over people
- Linear (in parameters)
 - $V(\text{bus}) = \beta_1 * (\text{walk time}) + \beta_2 * (\text{in-vehicle time}) + \beta_3 * (\text{cost/income}) + \dots$

$$U_{in} = V_{in} + \epsilon_{in}$$

Binary Logit Model

The diagram shows the Binary Logit Model equation for two alternatives, auto and transit. The equation is written as $V_{auto} = \beta_0 + \beta_1 TT_{auto} + \beta_2 age + \beta_3 Income$ and $V_{transit} = \beta_4 TT_{transit} + \beta_2 age + \beta_3 Income$. Annotations with arrows point to specific parts: 'alternative-specific constant (ASC)' points to β_0 ; 'attributes of alternatives' points to the TT terms ($\beta_1 TT_{auto}$ and $\beta_4 TT_{transit}$); and 'attributes of individual' points to the age and $Income$ terms.

$$\begin{aligned} V_{auto} &= \beta_0 + \beta_1 TT_{auto} + \beta_2 age + \beta_3 Income \\ V_{transit} &= \beta_4 TT_{transit} + \beta_2 age + \beta_3 Income \end{aligned}$$

alternative-specific constant (ASC)

attributes of alternatives

attributes of individual

$$age = \begin{cases} 1 & \text{if } age \geq 40 \\ 0 & \text{otherwise} \end{cases}$$

$$Income = \begin{cases} 1 & \text{if } income \geq 25K \\ 0 & \text{otherwise} \end{cases}$$

TT = travel time by a mode (auto, transit)

A simple example: Car or Bus?

$$P(auto) = \frac{e^{V_{auto}}}{e^{V_{auto}} + e^{V_{transit}}}$$

$$P(transit) = \frac{e^{V_{transit}}}{e^{V_{auto}} + e^{V_{transit}}}$$



A simple example: Car or Bus?

$$\begin{aligned}
 V_{auto} &= \beta_0 + \beta_1 TT_{auto} + \beta_2 age + \beta_3 Income \\
 V_{transit} &= \beta_4 TT_{transit}
 \end{aligned}$$

$m = \begin{cases} 1 & \text{if income} \geq 25K, \\ 0 & \text{otherwise.} \end{cases}$

$a = \begin{cases} 1 & \text{if age} \geq 40, \\ 0 & \text{otherwise,} \end{cases}$

	Constant (β_0)	TT ($\beta_1 \beta_4$)	Age (β_2)	Income (β_3)
Auto	0.0	-0.5	1	1
Transit	0.0	-0.8		

A simple example: Car or Bus?

$$a = 1, \quad m = 1, \quad TT_{\text{auto}} = 48, \quad TT_{\text{transit}} = 30,$$

A simple example: Car or Bus?

$$a = 1, \quad m = 1, \quad TT_{\text{auto}} = 48, \quad TT_{\text{transit}} = 30,$$

$$V_{\text{auto}} = -0.5 \cdot 48 + 1 \cdot 1 + 1 \cdot 1 = -24 + 2 = -22,$$

$$V_{\text{transit}} = -0.8 \cdot 30 + 1 \cdot 1 + 1 \cdot 1 = -24 + 2 = -22,$$

$$P(\text{auto}) = \frac{e^{V_{\text{auto}}}}{e^{V_{\text{auto}}} + e^{V_{\text{transit}}}},$$

$$P(\text{transit}) = 1 - P(\text{auto}).$$

A simple example: Car or Bus?

$$a = 1, \quad m = 1, \quad TT_{\text{auto}} = 48, \quad TT_{\text{transit}} = 30,$$

$$V_{\text{auto}} = -0.5 \cdot 48 + 1 \cdot 1 + 1 \cdot 1 = -24 + 2 = -22,$$

$$V_{\text{transit}} = -0.8 \cdot 30 + 1 \cdot 1 + 1 \cdot 1 = -24 + 2 = -22,$$

$$P(\text{auto}) = \frac{e^{-22}}{e^{-22} + e^{-22}} = \frac{1}{2},$$

$$P(\text{transit}) = \frac{e^{-22}}{e^{-22} + e^{-22}} = \frac{1}{2}.$$

What if we have more than 2 alternatives?

- **Multinomial logit extends binary logit**
 - One formula, any number of alternatives (e.g., auto, bus, subway, bike)
- **Independence of Irrelevant Alternative (IIA) limitation: choices must be independent**
 - Implies that the relative odds of two options are unaffected by a third (often unrealistic)

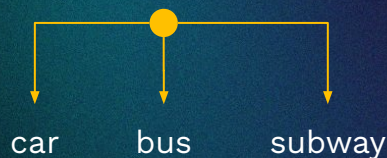
Multinomial logit

$$U_{auto} = V_{auto} + \varepsilon_{auto}$$

$$U_{bus} = V_{bus} + \varepsilon_{bus}$$

$$U_{rail} = V_{rail} + \varepsilon_{rail}$$

$$P(auto) = \frac{e^{V_{auto}}}{e^{V_{auto}} + e^{V_{bus}} + e^{V_{subway}}}$$



What if we have more than 2 alternatives?

- **Multinomial logit extends binary logit**
 - One formula, any number of alternatives (e.g., auto, bus, subway, bike)
- **IIA limitation: choices must be independent**
 - Implies that the relative odds of two options are unaffected by a third (often unrealistic)
- **Nested logit handles correlated options**
 - Groups “similar” alternatives into nests (e.g., bus + subway as “transit”) to relax IIA.

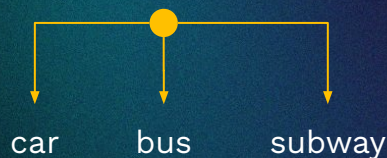
Multinomial logit

$$U_{auto} = V_{auto} + \varepsilon_{auto}$$

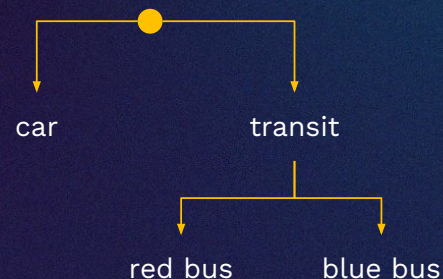
$$U_{bus} = V_{bus} + \varepsilon_{bus}$$

$$U_{rail} = V_{rail} + \varepsilon_{rail}$$

$$P(auto) = \frac{e^{V_{auto}}}{e^{V_{auto}} + e^{V_{bus}} + e^{V_{subway}}}$$



Nested logit



AI for Transportation

Acknowledgements:
Shenhao Wang

R2: Discrete Choice Modeling

S1: Refresher on Discrete Choice Modeling

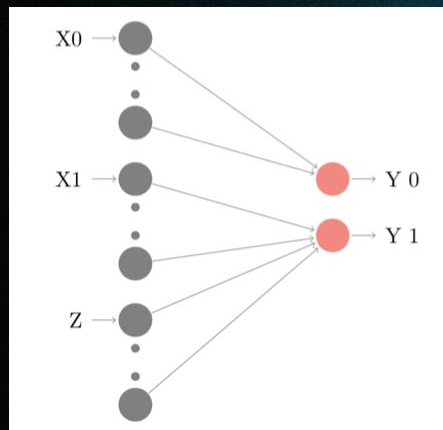
S2: Leveraging ASU-DNN

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Discussed models

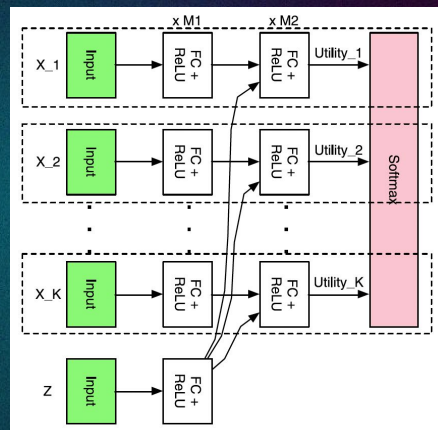
MNL



$$V_k = f(x_k, z) = \beta^\top x_k + \gamma^\top z$$

Alternative-Specific
Utility Functions

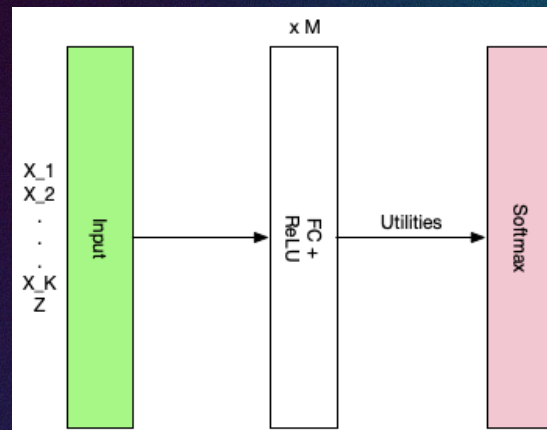
ASU-DNN



$$V_k = f(x_k, z)$$

Fully Connected

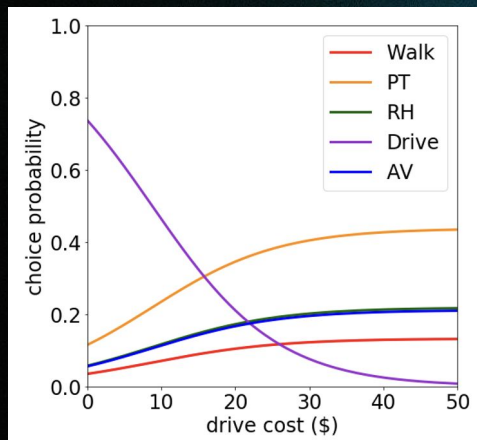
FC-DNN



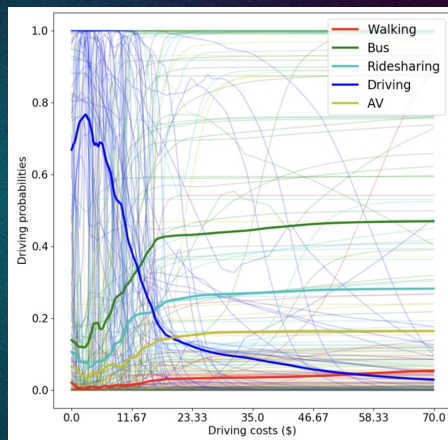
$$V_k = f(x_1, x_2, \dots, x_K, z)$$

Discussed results

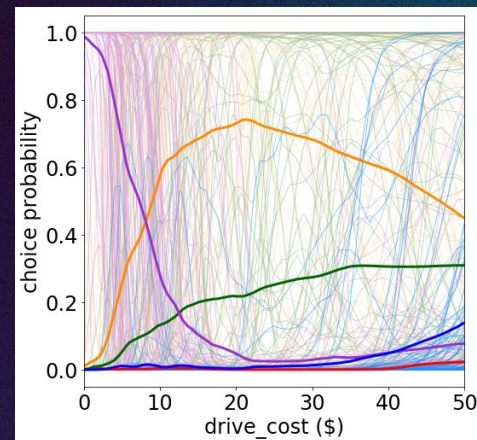
MNL



ASU-DNN



FC-DNN



Comparing classical and NN models

	Discrete Choice	Machine Learning
Driven by	theory	data
Captures	causality	correlation
Focus on	interpretability	accuracy
Model	choice model	classifier
Model selection	hypothesis testing	out-of-sample validation
Main usage	what-if analysis	interpolation

Exercises

Model commuter mode choice in the Netherlands (train vs. car) using 1987 survey data.

- Methods:
 - Logistic Regression
 - Multinomial Logit
 - Fully Connected Deep Neural Network
 - ASU Deep Neural Network

What we will be doing

Model commuter mode choice in the Netherlands (train vs. car) using 1987 survey data.

- Methods:
 - Logistic Regression
 - Multinomial Logit
 - Fully Connected Deep Neural Network
 - ASU Deep Neural Network
- } **Baselines**

What we will be doing

Model commuter mode choice in the Netherlands (train vs. car) using 1987 survey data.

- Methods:
 - Logistic Regression
 - Multinomial Logit
 - Fully Connected Deep Neural Network
 - ASU Deep Neural Network
- Compare the results

A peek at the data

Variable	Description
id	individual identifier
choiceid	choice identifier
choice	one of choice1, choice2
pricez	price of proposition z ($z = 1, 2$) in cents of guilders
timez	travel time of proposition z ($z = 1, 2$) in minutes
comfortz	comfort of proposition z ($z = 1, 2$), values 0, 1, or 2 (decreasing comfort)
changez	number of changes for proposition z ($z = 1, 2$)

See it in action



AI for Transportation

R2: Discrete Choice Modeling

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Resulting parameters (MNL)

```
=====
MULTINOMIAL LOGIT (MNL) MODEL SUMMARY
=====
```

```
Log-Likelihood = -1723.837
```

```
AIC = 3457.67 | BIC = 3491.05
```

```
-----
Parameters:
```

```
Parameter  Estimate
```

```
ASC[j=2] -0.032498
```

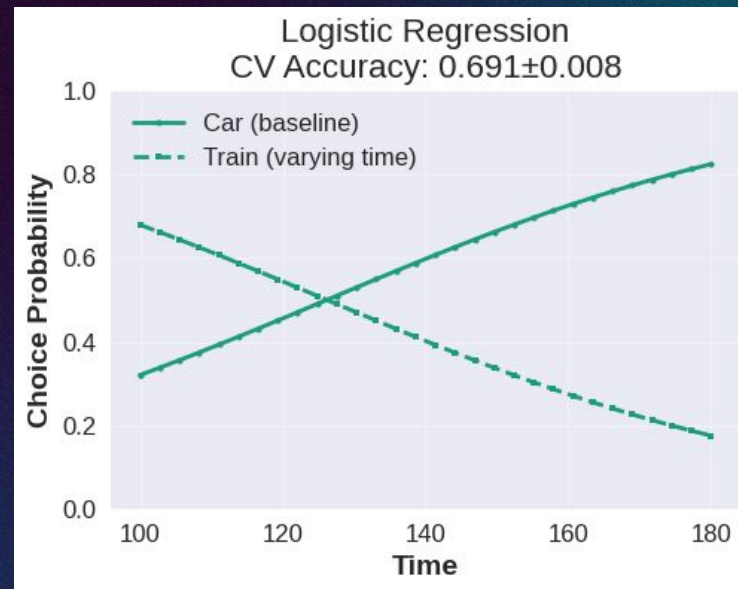
```
price -1.908640
```

```
time -0.820926
```

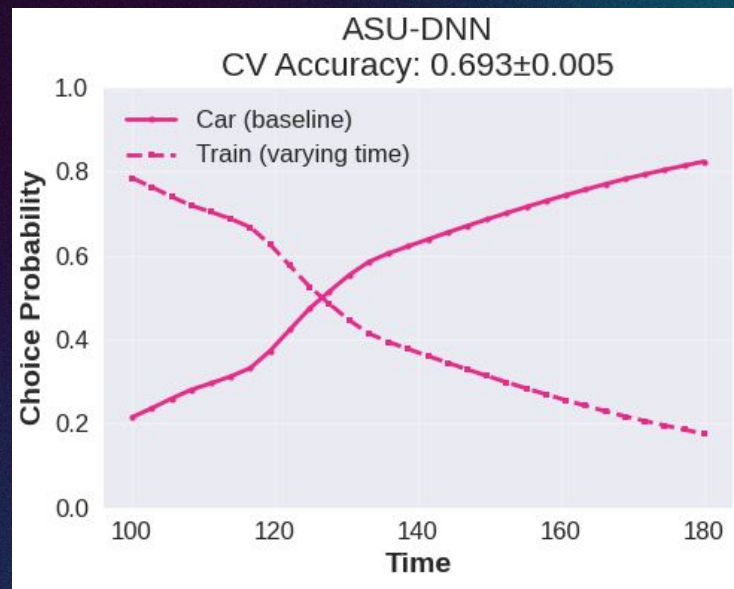
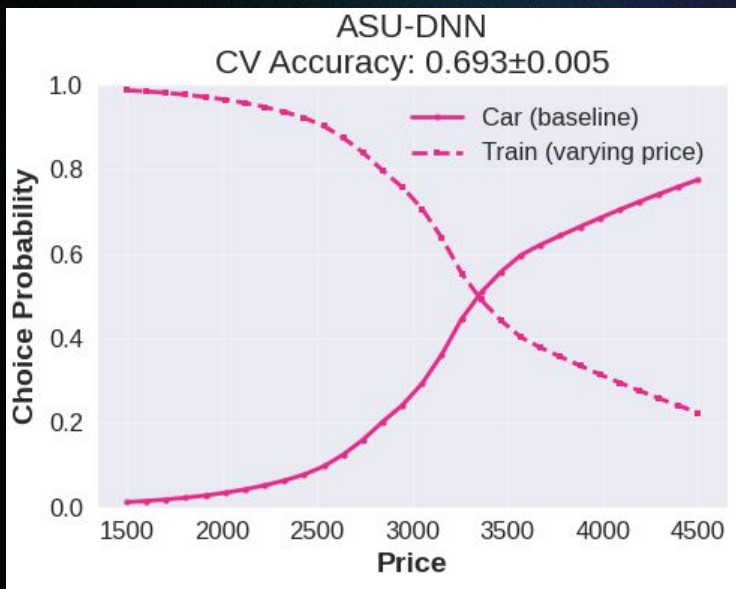
```
comfort -0.576485
```

```
change -0.240370
```


Resulting choice probabilities (MNL)



Resulting choice probabilities (ASU-DNN)



AI for Transportation

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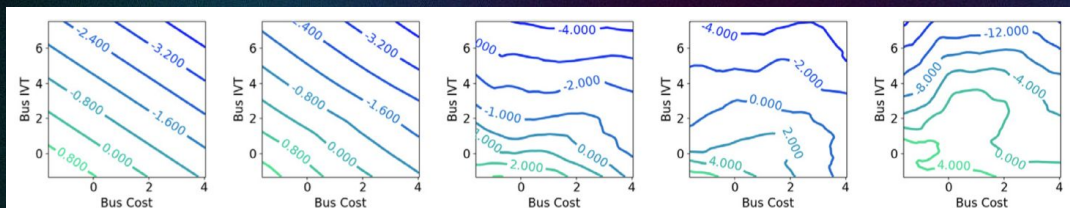
S3: Understanding the Results

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Theory-Based Hybrid Models

- **Theory-Based Residual Neural Network**

- Allow the interpolation between theory- and data-driven models



- **Neural-Embedded Taste Models (TasteNet-MNL):**

- Learns flexible, individual-specific taste parameters with a neural network
- Ensures interpretable outputs (e.g., Value of Time, elasticities) remain realistic

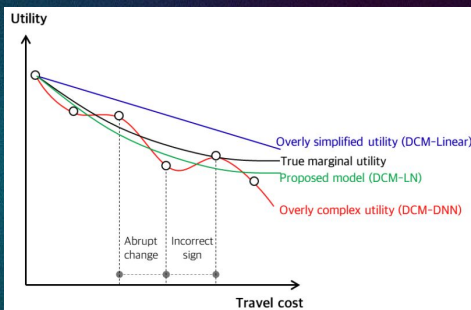
- **Graph Neural Nets for Network Effects:**

- Capture peer or social influence in choice behavior
- Encode private vs. social utilities through graph convolutions

Behavioral Regularization

- **Lattice Networks (Monotonic DCM):**

- Neural architecture with built-in monotonicity constraints
- Guarantees utility decreases with higher cost or time
- Produces stable and interpretable WTP and elasticity estimates



- **Domain-Knowledge Constraints:**

- Injects behavioral rules into DNN training via pseudo-observations or penalties
- Ensures substitution patterns and signs match economic intuition
- Balances flexibility of deep models with econometric consistency

Further Examples

- **Nest-GNN (Zhou et al., 2025):**
 - Alternative graphs for inter-alternative correlations, enabling message passing between related choices
 - Generalizes MNL, NL, and ASU-DNN within one framework, retaining the two-layer substitution patterns of Nested Logit
- **TB-ResNet (Wang et al., TRB 2021):**
 - Blends residual deep nets with classic logit
 - Stabilizes utility functions and gradients, boosting interpretability
- **SHAP-Enhanced NN (Tang et al., 2024):**
 - Neural model for mode choice with post-hoc SHAP explanations
 - Identified cost, car ownership, and income as top drivers of travel choice

